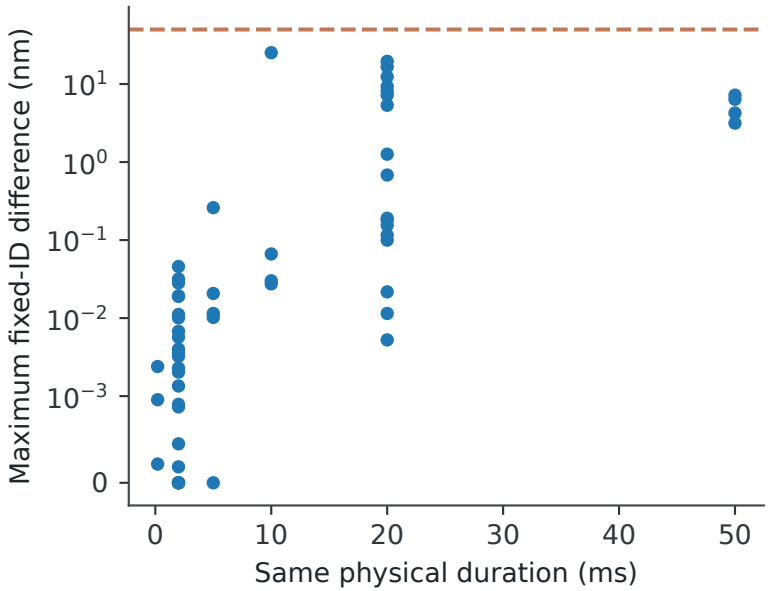
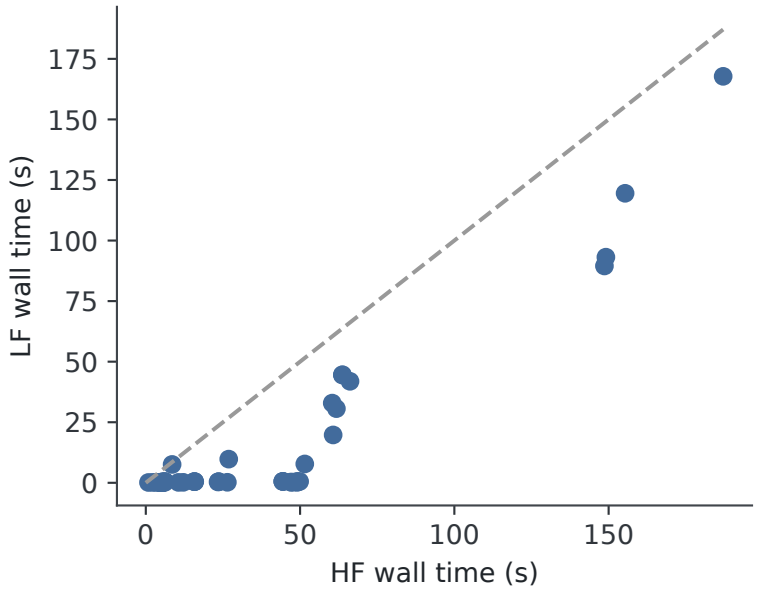
Paired slow-time validation

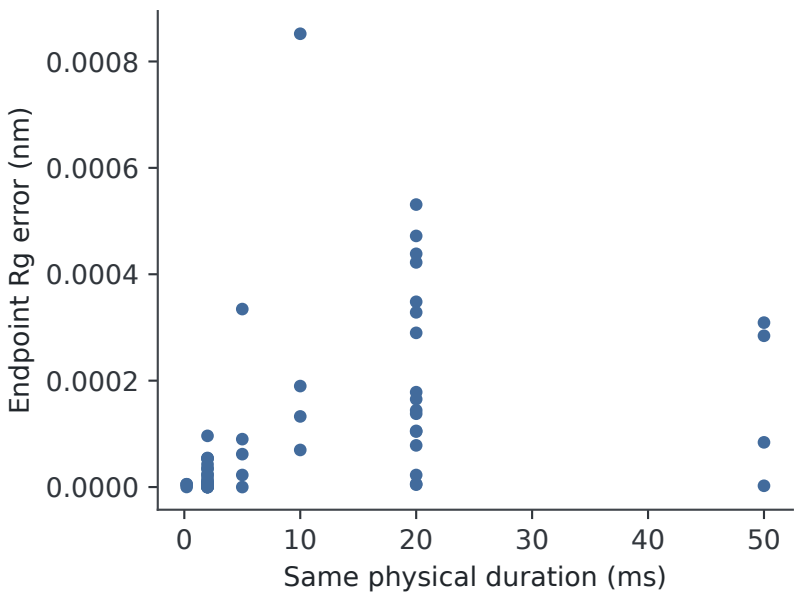


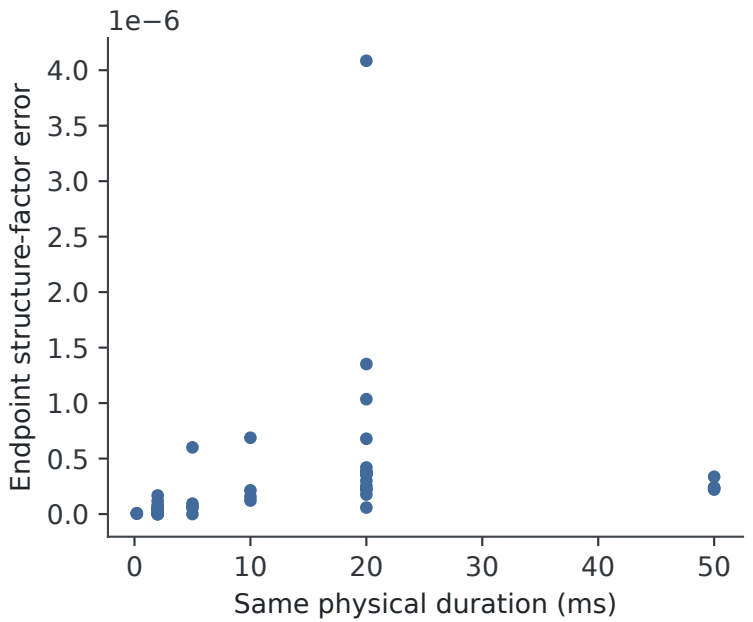


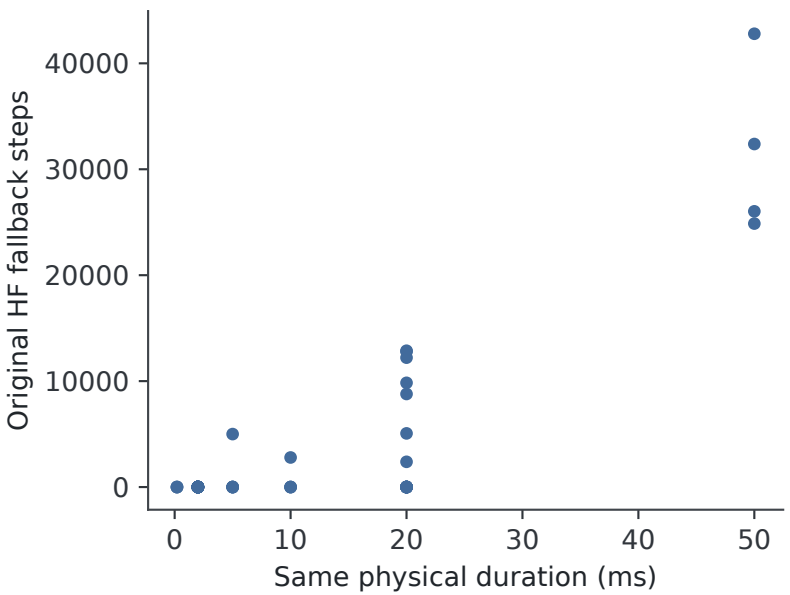
All original F1 and F2 retained
Far resistance: exactly A = I
Adaptive SciPy RK45
Conservative geometric interval guard
Original solver near contact

Compact failure reproduced at 92 us
No relaxed overlap / wall limits

LF scope is action- and time-specific.







Symmetric-log error axis: linear below 0.001 nm; exact zeros retained

EXPLORATORY | Homogeneous water | No flow or streaming | Uncalibrated physics and electrical mapping